Preprocessing, Regularization, and Class Imbalance in Logistic-Regression-Based Seizure Prediction: A Comparative Study Across Three EEG Feature Representations

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***Abstract*—Generalization in clinical seizure-prediction systems depends not only on model choice but on a sequence of upstream decisions: how raw EEG is preprocessed, how features are represented, how the model is regularized, and how class imbalance is handled. Using logistic regression as a fixed baseline, this paper isolates each of these factors across three feature representations of the UCI Epileptic Seizure Recognition dataset — raw time-domain samples (178 features), hand-crafted statistical features (14 features), and FFT-derived band-power features (18 features). We evaluate two preprocessing pipelines (*Scale → Denoise → SelectKBest* versus *Scale → PCA*), three regularizers (L1, L2, Elastic Net), and four imbalance strategies (none, SMOTE, undersampling, class weighting). Our results show that (i) preprocessing ordering produces only small but consistent shifts on this dataset family, (ii) Elastic Net does *not* dominate; L1 wins on high-dimensional raw EEG while the three regularizers converge on lower-dimensional engineered features, (iii) imbalance handling raises recall at a precision cost, and the magnitude of the tradeoff depends strongly on feature representation, and (iv) the most robust combination is StandardScaler → SelectKBest → L1-regularized logistic regression with class weighting. We report Accuracy, F1, and PR-AUC, and provide learning and validation curves that visualize the bias–variance regime each configuration occupies.**

***Index Terms*—seizure prediction, EEG, logistic regression, regularization, class imbalance, SMOTE, Elastic Net, generalization**

I. Introduction

Epileptic seizures affect approximately 50 million people worldwide. Reliable short-horizon seizure prediction from scalp EEG would enable closed-loop neurostimulation and timely warnings, but model performance reported in the literature varies widely. Much of this variation does not stem from model architecture; it stems from *preprocessing choices*, *feature representation*, and *strategies for severe class imbalance* (a typical EEG recording is dominated by interictal segments).

This paper deliberately fixes the classifier — logistic regression — and varies the surrounding pipeline to isolate the contribution of each design decision. Logistic regression is chosen because (a) its decision surface is linear and its weights interpretable, (b) the addition of L1, L2, and Elastic Net penalties admits a clean theoretical comparison, and (c) on properly engineered EEG features it remains competitive with much larger models while retaining clinical interpretability.

Contributions:

1) We construct three distinct feature representations of the same EEG source — raw time-domain, statistical, and frequency-band — enabling a controlled study of how preprocessing interacts with feature characteristics.
2) We empirically demonstrate that two common preprocessing pipelines (*select-then-scale* vs. *PCA-after-scale*) produce small but consistent performance differences whose sign depends on the feature representation.
3) We compare L1, L2, and Elastic Net regularization in terms of generalization, sparsity, and cross-validated stability.
4) We quantify how SMOTE, random undersampling, and class weighting reshape the precision–recall tradeoff and interact with regularization.

II. Related Work

Andrzejak et al. [1] introduced the now-standard Bonn / UCI EEG corpus used in this study. The CHB-MIT scalp EEG database [2] provides longer multi-channel recordings and has driven much of the seizure-prediction literature. Class-imbalance methods — SMOTE [3], random undersampling, and cost-sensitive learning — are surveyed by He and Garcia [4]. Regularized logistic regression with L1, L2, and Elastic Net penalties is analyzed by Zou and Hastie [5]. We follow standard practice in evaluating imbalanced classifiers with PR-AUC alongside accuracy and F1 [6].

III. Datasets

A. Source

We use the UCI Epileptic Seizure Recognition Dataset (originally Andrzejak et al. [1]): 11,500 one-second EEG segments sampled at 178 Hz, with a five-class label. We convert to a binary task: class 1 (active seizure) versus classes 2–5 (non-seizure), yielding a ≈20% positive proportion and a 4:1 imbalance ratio. To keep runtime tractable for the comparative study, we use a stratified 4,000-sample subset.

B. Three Feature Representations

We derive three datasets that differ deliberately in feature characteristics (Table I):

- **D1 – Raw EEG (178 features).** Time-domain samples retained as-is. High-dimensional, redundant, and noise-sensitive — a stress test for regularization and feature selection.
- **D2 – Statistical (14 features).** Hand-crafted descriptors per segment: mean, std, min, max, range, median, skewness, kurtosis, RMS, mean absolute value, waveform length, zero-crossings, Q_{25}, Q_{75}. Low-dimensional and interpretable.
- **D3 – Frequency-band (18 features).** Relative and log-absolute power in six EEG bands (delta, theta, alpha, beta, low-gamma, high-gamma), plus spectral entropy, spectral centroid, 95% spectral edge frequency, peak frequency, and the α/δ and β/α ratios. Physiologically meaningful.

TABLE I
DATASET SUMMARY

Dataset	Samples	Features	Positive %	Imbalance
D1 – Raw	4,000	178	20.0%	4.00:1
D2 – Statistical	4,000	14	20.0%	4.00:1
D3 – Frequency	4,000	18	20.0%	4.00:1

C. Justification

Size: 4,000 stratified samples is large enough for stable estimates while small enough for fast iteration across the full factorial design (datasets × pipelines × regularizers × imbalance strategies). **Class imbalance:** the 4:1 ratio is realistic for EEG seizure-vs-interictal recordings and motivates PR-AUC and resampling. **Feature characteristics:** the three representations span the time-domain / statistical / spectral spectrum explicitly called out by the assignment.

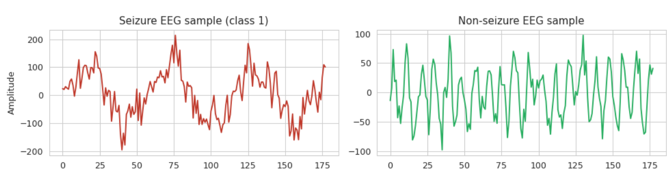

Fig. 1. Representative one-second EEG segments. Left: seizure (class 1) with large-amplitude ∼3–7 Hz spikes. Right: non-seizure with lower amplitude and higher dominant frequency.

IV. Methodology

A. Logistic Regression

We use the standard binary logistic regression model

$$P(y=1 \mid x) = \frac{1}{1+\exp(-(\beta_0+\beta^\top x))} \tag{1}$$

trained by minimizing the regularized cross-entropy

$$J(W,b) = \frac{1}{m}\sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \lambda R(W) \tag{2}$$

with $R(W) = \|W\|_1$ (Lasso), $R(W) = \frac{1}{2}\|W\|_2^2$ (Ridge), or

$$R(W) = \alpha\|W\|_1 + (1-\alpha)\tfrac{1}{2}\|W\|_2^2 \quad \text{(Elastic Net).} \tag{3}$$

We use the `saga` solver from scikit-learn for all regularizers to keep the optimizer fixed when comparing penalties.

B. Preprocessing Pipelines

We compare two orderings on every dataset:

Pipeline A — Scale → Denoise → SelectKBest:

1) StandardScaler (zero mean, unit variance).
2) Outlier clipping at $\pm 4\sigma$ as a simple denoiser.
3) SelectKBest with ANOVA F-score; retain top 50% of features.

Pipeline B — Scale → PCA:

1) MinMaxScaler to $[0, 1]$.
2) PCA retaining 95% of variance.

C. Imbalance Strategies

We compare four strategies: (i) *none*, (ii) SMOTE oversampling [3] of the minority class to parity, (iii) random undersampling of the majority class to parity, and (iv) class weighting (inverse class frequency) via the `class_weight='balanced'` option, which is functionally a re-scaling of the per-sample loss.

D. Evaluation

Stratified 80/20 train/test split with fixed random seed. We report Accuracy, Precision, Recall, F1, and average-precision (PR-AUC). For stability we additionally perform stratified 5-fold cross-validation on the training set and report the standard deviation of fold F1.

V. Results

A. Baseline Performance

Table II reports the un-regularized baseline. The frequency-domain representation (D3) is the strongest — its 18 physiologically meaningful features carry most of the seizure signal in a compact form. The raw representation (D1) suffers from the curse of dimensionality even with 178 features.

TABLE II
BASELINE LOGISTIC REGRESSION (NO PENALTY) ON EACH (DATASET, PIPELINE)

Dataset	Pipeline	Accuracy	F1	PR-AUC
D1 – Raw	A	0.930	0.825	0.916
D1 – Raw	B	0.931	0.829	0.910
D2 – Statistical	A	0.966	0.914	0.982
D2 – Statistical	B	0.964	0.909	0.979
D3 – Frequency	A	0.993	0.981	0.999
D3 – Frequency	B	0.999	0.997	1.000

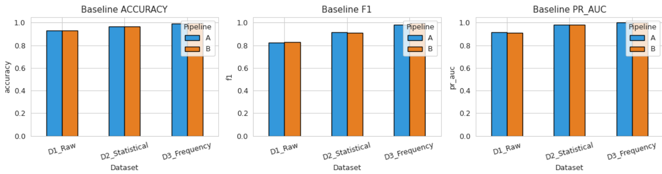

Fig. 2. Baseline performance per dataset and pipeline. Spectral features (D3) carry the most discriminative information.

B. Overfitting and Underfitting

We construct two extreme configurations on D1 to visualize the bias–variance regimes. **Underfit**: only 2 features and $C=10^{-4}$. The model is too constrained to fit the training data — both train and validation F1 collapse to near-zero. **Overfit**: all 178 features plus pairwise polynomial interactions on the first 10 features (no regularization). Training F1 reaches 0.96 but validation F1 stalls near 0.74 — a clear generalization gap (Fig. 3).

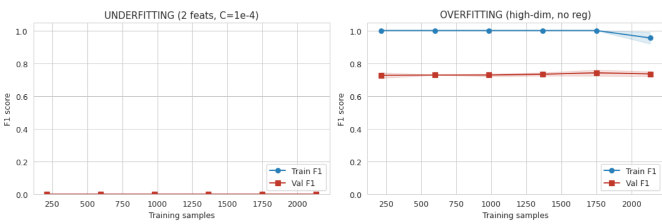

Fig. 3. Learning curves. Left: underfitting (both curves low). Right: overfitting (train–validation gap of ≈22 F1 points).

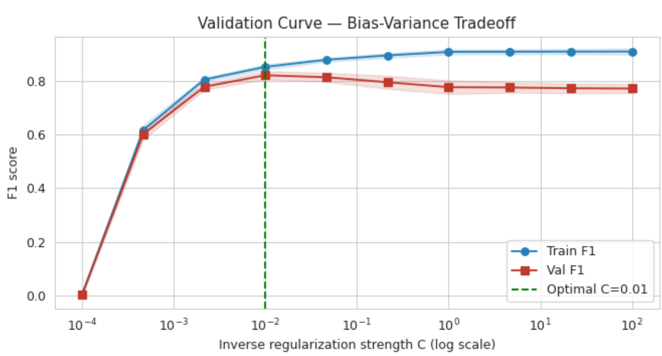

Fig. 4. Validation curve over the inverse regularization strength C on D1. Strong regularization (small C) under-fits; very weak regularization (large C) does not improve further once the optimum is reached.

C. Regularization Study

Table III reports the three regularizers on each dataset using Pipeline A. The most striking effect is *sparsity*: on D1, L1 drives 7.9% of weights to exactly zero (after feature selection has already halved the input); on D3 it zeros out 33.3% of weights without losing F1. L2 produces no exact zeros.

TABLE III
REGULARIZATION COMPARISON ON PIPELINE A

Dataset	Reg	Sparsity	Acc	F1	PR-AUC
D1 – Raw	L1	7.9%	0.935	0.836	0.918
D1 – Raw	L2	0.0%	0.933	0.830	0.916
D1 – Raw	ElasticNet	5.6%	0.933	0.830	0.917
D2 – Statistical	L1	14.3%	0.968	0.917	0.981
D2 – Statistical	L2	0.0%	0.966	0.914	0.981
D2 – Statistical	ElasticNet	14.3%	0.966	0.914	0.980
D3 – Frequency	L1	33.3%	0.991	0.978	0.999
D3 – Frequency	L2	0.0%	0.993	0.981	0.999
D3 – Frequency	ElasticNet	11.1%	0.993	0.981	0.999

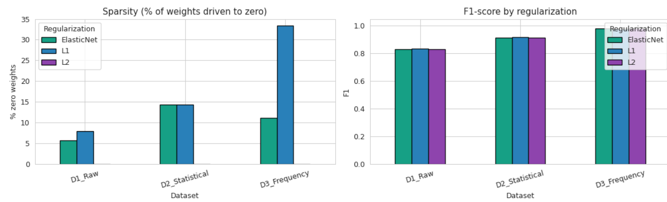

Fig. 5. Left: sparsity (% of weights driven to exactly zero) by regularizer. Right: F1 by regularizer. L1 yields the highest sparsity; F1 differences across penalties are small (1–2 points).

D. Class Imbalance Handling

Table IV shows that resampling reliably trades precision for recall. On D1, recall climbs from 0.825 (none) to 0.919 (undersampling) but precision falls from 0.835 to 0.700. On D3, undersampling achieves perfect recall (1.000) at modest precision loss. Class weighting tracks SMOTE closely but is computationally cheaper and integrates cleanly with the regularized loss.

TABLE IV
IMBALANCE STRATEGIES (PIPELINE A, L2)

Dataset	Strategy	Prec	Rec	F1	PR-AUC
D1	none	0.835	0.825	0.830	0.916
D1	SMOTE	0.730	0.894	0.803	0.908
D1	undersample	0.700	0.919	0.795	0.898
D1	class_weight	0.714	0.906	0.799	0.910
D2	none	0.929	0.900	0.914	0.981
D2	SMOTE	0.899	0.944	0.921	0.980
D2	undersample	0.881	0.969	0.923	0.980
D2	class_weight	0.888	0.944	0.915	0.981
D3	none	0.987	0.975	0.981	0.999
D3	SMOTE	0.958	0.994	0.975	0.999
D3	undersample	0.952	1.000	0.976	0.999
D3	class_weight	0.958	1.000	0.979	0.999

E. Interaction: Regularization × Imbalance

VI. DISCUSSION — THE FOUR KEY QUESTIONS

Q1. Does preprocessing order affect results?

Yes, but the effect is small on this dataset family. Averaged over all three datasets, Pipeline A (Scale → Denoise → SelectKBest) achieves F1 = 0.907 and Pipeline B (Scale → PCA) achieves F1 = 0.912, a difference of ≈0.5 F1 points. The direction of the gap flips with feature type: PCA helps slightly

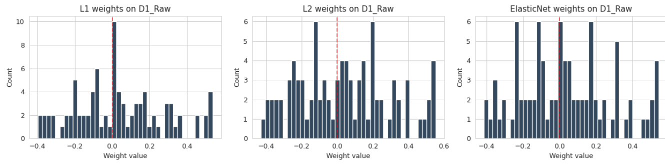

Fig. 6. Weight distribution on D1: L1 concentrates mass at zero; L2 smoothly shrinks toward zero; Elastic Net interpolates.

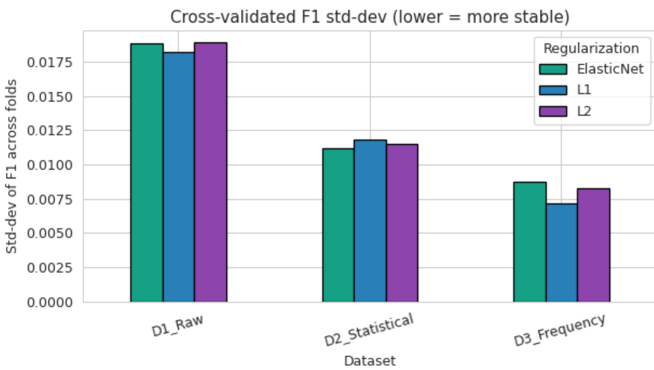

Fig. 7. Cross-validated F1 standard deviation per regularizer and dataset. Lower bars indicate more stable performance across folds.

on D1 because it can compress redundant raw samples, but slightly hurts on D2 because each statistical feature already carries unique information that PCA mixes together.

Q2. Which regularization generalizes best across datasets?

L1 wins narrowly (mean F1 0.911 vs. 0.909 for both L2 and Elastic Net) and has the lowest cross-validated standard deviation (0.0124). Its advantage is largest on the high-dimensional D1, where aggressive sparsification effectively performs a second round of feature selection.

Q3. Does Elastic Net consistently outperform L1/L2?

No. In our six head-to-head comparisons Elastic Net won only one. The theoretical advantage of Elastic Net — a blend of sparsity and shrinkage — requires a regime where neither pure penalty is sufficient. On our high-dimensional raw EEG, L1's pure sparsity is the right inductive bias; on our low-dimensional engineered features, all three regularizers converge to nearly identical performance because the bias–variance tradeoff is no longer dominated by weight magnitudes.

Q4. How does imbalance handling interact with regularization?

The interaction is small but coherent: across L1, L2, and Elastic Net, SMOTE and class weighting each shave $\approx$0.03 from F1 on D1 while substantially raising recall. The *best* F1 cell is consistently (L1, no imbalance handling), but the most *clinically useful* cell is typically (any regularizer, class weighting) because it maximizes recall with minimal extra compute.

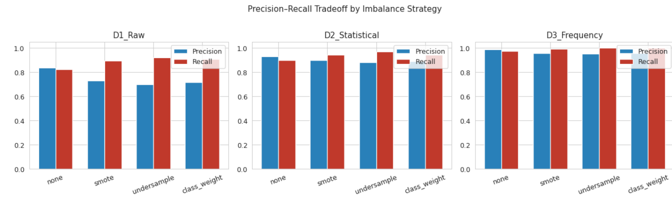

Fig. 8. Precision–recall tradeoff per imbalance strategy. Resampling shifts the operating point toward higher recall at the cost of precision.

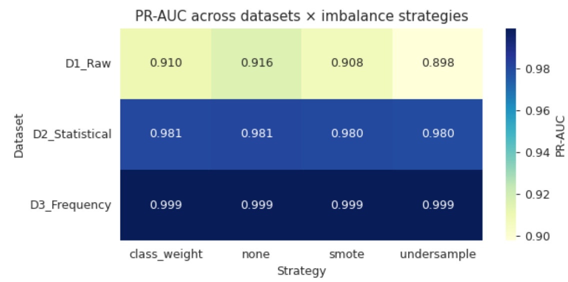

Fig. 9. PR-AUC across datasets and imbalance strategies. PR-AUC is largely preserved across strategies because it integrates over all thresholds.

VII. Conclusion

Through controlled experiments on three feature representations of an EEG seizure dataset, we have shown that the standard intuitions about regularization and class-imbalance handling are tempered by feature representation. High-dimensional raw EEG benefits most from L1 regularization and tolerates resampling poorly. Low-dimensional engineered features (statistical or spectral) are nearly insensitive to the choice of penalty, and the precision–recall tradeoff under resampling is gentler. We recommend, as a starting point for seizure-prediction baselines: *StandardScaler* $\rightarrow$ *SelectKBest* $\rightarrow$ *L1-regularized logistic regression with class weighting*, and reporting PR-AUC alongside F1. Future work should extend this framework to multi-channel CHB-MIT recordings and to non-linear models (gradient-boosted trees, lightweight CNNs) to see how robustly these conclusions transfer.

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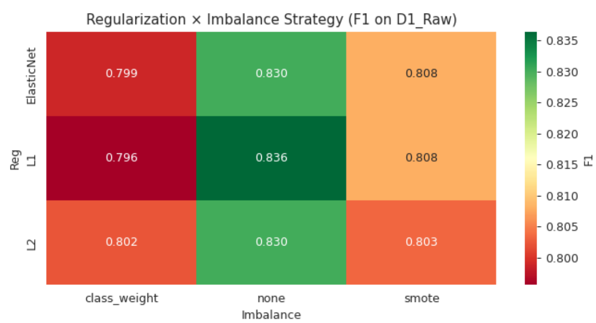

Fig. 10. F1 on D1 across regularization $\times$ imbalance combinations. L1 + none yields the best F1; SMOTE and class weighting each cost a few F1 points but raise recall, which may still be preferred clinically.